

1 / 5 | Ages 9-12 | 20 minutes

## The river crossing plan

### What we are practising

Plan forward and backward moves while checking every intermediate state.

### What you need

Paper / Pencil / Four large paper labels

Draw the riverbanks and write positions instead of moving cards.

### Before you begin

Use a farmer, fox, chicken and grain. The boat carries the farmer plus at most one other item.

This is a paper logic story only; do not approach water or involve real animals.

### Try it together

1. Start all four on the left bank; only the farmer operates the boat.
2. Do not leave fox with chicken, or chicken with grain, on a bank without the farmer.
3. Try crossings and record both banks after every move.
4. Continue until everyone reaches the right bank safely within the puzzle rules.

### Talk, draw or show your idea

Why might carrying something back help the plan succeed?

### Space to explore

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2 / 5 | Ages 9-12 | 15 minutes

## Only one label is true

### What we are practising

Test candidate solutions against an exact truth condition.

### What you need

Paper / Pencil

Draw labelled circles instead of boxes.

### Before you begin

Draw red, blue and green boxes; exactly one contains a prize and exactly one label is true.

Use names as well as colours; no hidden real prize or reward is needed.

### Try it together

1. Write red label: The prize is in blue. Blue label: The prize is not here. Green label: The prize is not in red.
2. Assume the prize is in red and mark all three labels true or false.
3. Repeat assuming blue, then green.
4. Choose the location producing exactly one true label.

### Talk, draw or show your idea

Why must you test every label for each possible location?

### Space to explore

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3 / 5 | Ages 9-12 | 20 minutes

## The shortest delivery route

### What we are practising

Compare route totals instead of judging by drawing distance.

### What you need

Paper / Pencil

Write the possible connections in a table instead of drawing a network.

### Before you begin

Draw points A, B, C, D, E and connect  $AB=2$ ,  $AC=5$ ,  $BC=1$ ,  $BD=4$ ,  $CD=1$ ,  $CE=6$ ,  $DE=3$ . All links work both ways.

All routes are fictional; do not use the drawing for real travel.

### Try it together

1. Find two routes from A to E and add their labelled costs.
2. Compare reaching C directly with reaching it through B.
3. Use the cheaper intermediate choices to find a better route.
4. Check the final total against the other routes you tried.

### Talk, draw or show your idea

Can a route with more links still have a lower total cost?

### Space to explore

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4 / 5 | Ages 9-12 | 15 minutes

## Crack the four-digit order

### What we are practising

Use adjacency and ordering constraints to eliminate permutations.

### What you need

Paper / Pencil

Use large handwritten cards for digits 1, 2, 3 and 4.

### Before you begin

The code uses 1, 2, 3 and 4 exactly once, read from left to right.

Use an invented puzzle code, never a real password or device PIN.

### Try it together

1. Write the clues: 1 is immediately before 3; 4 is before 2; 2 is not last.
2. Treat 1 and 3 as an ordered block and try possible placements.
3. Reject any order that breaks one clue.
4. Check the remaining code against all three clues.

### Talk, draw or show your idea

How does treating 13 as a block reduce the search?

### Space to explore

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5 / 5 | Ages 9-12 | 20 minutes

## Thirteen-token strategy

### What we are practising

Reason backward from a target and maintain a useful total.

### What you need

13 large paper tokens / Paper / Pencil

Draw 13 circles and cross them out.

### Before you begin

Two players alternate taking 1, 2 or 3 tokens; whoever takes the last token wins.

Keep competition friendly and use large tokens around younger children.

### Try it together

1. Play once from 13 tokens and record each move.
2. Study what happens when a player starts a turn with 4 tokens.
3. Look for replies that make the two players' moves total 4.
4. Start again and test taking 1 first, then completing each pair to 4.

### Talk, draw or show your idea

Why is leaving 4 tokens useful just before the other player's turn?

### Space to explore

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## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 1. The river crossing plan

#### A little hint

Take the chicken first; later you will need to bring it back once.

#### Answer / what to notice

One solution: take chicken across; return alone; take fox across; bring chicken back; take grain across; return alone; take chicken across.

Bringing the chicken back prevents either forbidden pair from being left together without the farmer.

#### Make it easier

Provide the first two crossings and ask for the next one.

#### Take it further

Find the alternative solution that transports grain before fox.

#### Another day, away from the screen

Replace the characters with invented robots that cannot be left in certain pairs.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 2. Only one label is true

#### A little hint

If the prize is in red, the blue label says something true.

#### Answer / what to notice

The prize is in red: red label false, blue true, green false. Blue or green each makes two labels true.

Only the red location satisfies both one prize and exactly one true label.

#### Make it easier

Use a three-row table with an adult reading the labels aloud.

#### Take it further

Change the condition to exactly two true labels and explain why there are two possible locations.

#### Another day, away from the screen

Invent a two-box puzzle and check whether its clues produce zero, one or several solutions.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 3. The shortest delivery route

#### A little hint

A to B to C costs 3, less than the direct A to C cost of 5.

#### Answer / what to notice

The shortest route is A-B-C-D-E with cost  $2+1+1+3=7$ .

The cheapest costs from A are B=2, C=3, D=4 and E=7; the diagram's drawn distances are irrelevant.

#### Make it easier

Compare only A-B-D-E and A-B-C-D-E.

#### Take it further

Close the C-D link and find the equally best routes. Both A-B-D-E and A-B-C-E cost 9.

#### Another day, away from the screen

Design a five-stop network with two equally cheap routes.



## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 4. Crack the four-digit order

#### A little hint

Arrange the three pieces 13, 4 and 2; then apply the other two clues.

#### Answer / what to notice

The code is 4213, read left to right: four, two, one, three.

With 4 before 2, the possible block orders are 13-4-2, 4-13-2 and 4-2-13; only the last avoids ending in 2.

#### Make it easier

Provide the three possible block orders to compare.

#### Take it further

Remove one clue and count how many codes become possible.

#### Another day, away from the screen

Write an ordering puzzle with four animal names and verify its solution count.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 5. Thirteen-token strategy

#### A little hint

If the other player takes 1, take 3; if they take 2, take 2; if they take 3, take 1.

#### Answer / what to notice

The first player can take 1, leaving 12, then reply so each pair removes 4, leaving 8, 4 and finally 0.

Keeping the opponent's starting total at a multiple of 4 forces the final winning reply when the rule is followed.

#### Make it easier

Start with 5 tokens and test the same first move.

#### Take it further

Change the maximum take to 4 and investigate a useful starting total.

#### Another day, away from the screen

Teach the strategy to a partner, then cooperate to explain why it works.